

InsightLens: A Vision-Language Based Accessibility Framework for Extracting and Understanding Classroom Content in Dual Languages

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Abstract

The rapid development of large language models (LLMs) has transformed educational technologies, but processing **Banglish** (Bengali + English code-mixed) lecture videos remains challenging. This thesis presents a **multimodal AI pipeline** that processes both **audio (speech)** and **visual (whiteboard)** content to automatically generate comprehensive lecture notes. The system employs Whisper-large-v3-turbo for Banglish ASR, BanglaASR for pure Bengali segments, and Qwen2.5-VL-7B for whiteboard text extraction. Evaluated on 9 real classroom videos across Python, Digital Logic Design, and DBMS domains, the system achieves **73.9% Term F1 score, 83.6% Precision, and 66.5% Recall**. A critical finding shows that visual bias in ASR—while theoretically promising—causes hallucinations in practice and was disabled in production. This work contributes the **first ground-truth Banglish lecture dataset**, a **term-based evaluation framework for code-mixed languages**, and a **complete end-to-end multimodal processing pipeline**.

Problem Statement

Processing Banglish (Bengali + English code-mixed) lecture videos is difficult because:

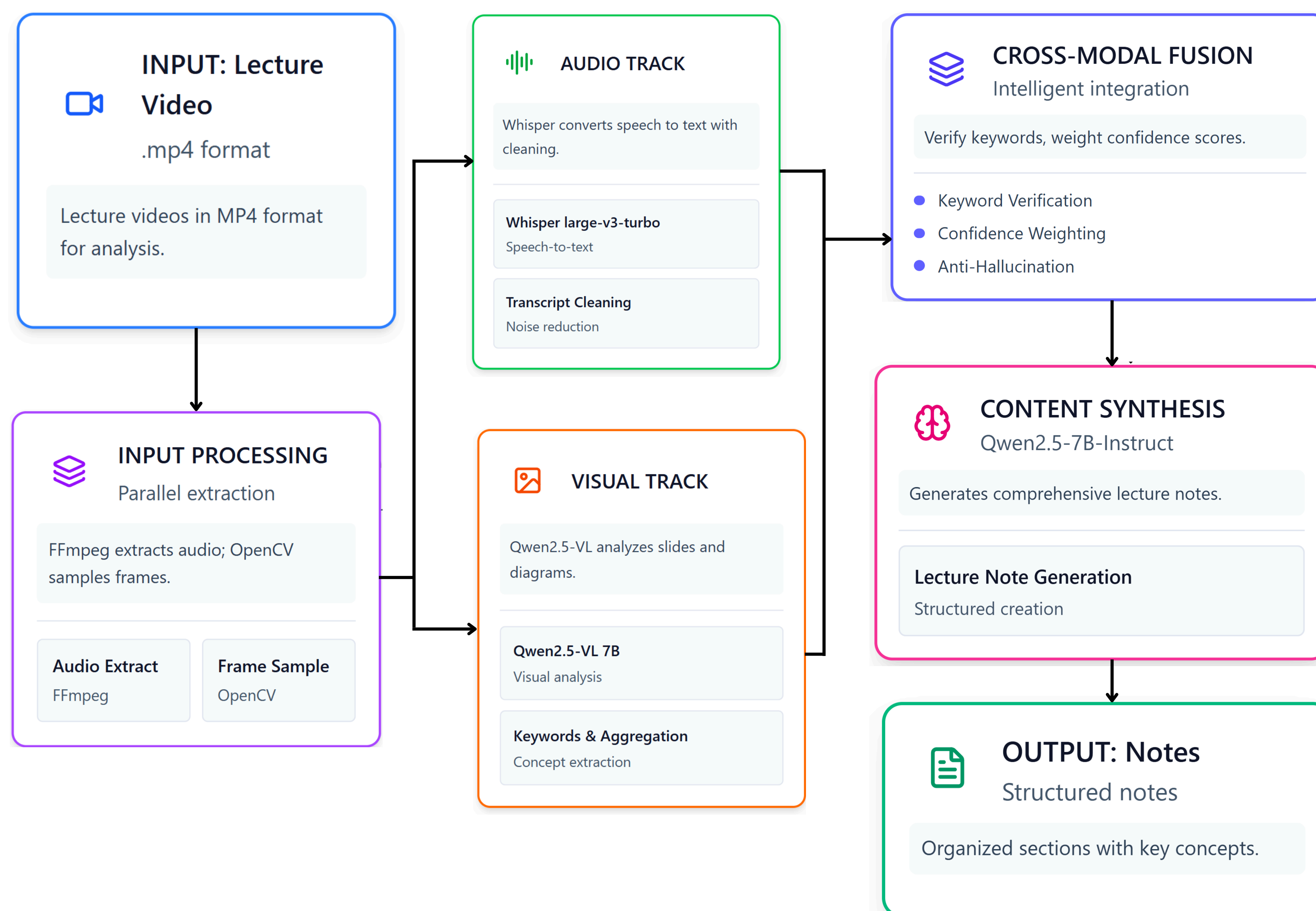
- **Code-mixing**: Speakers switch between Bengali and English mid-sentence
- **Technical terms** spoken with Bengali grammar markers (e.g., "class-ta", "object-ke")
- **ASR hallucinations**: Whisper misinterprets technical terms or produces repetition loops
- **Unicode vs Romanized mismatch**: BanglaASR outputs Bengali Unicode (আমি), Whisper outputs Romanized text (ami)
- **No standard Romanization**: Multiple valid spellings exist—traditional WER is inappropriate
- **Whiteboard ground truth**: Critical technical content visible but requires VLM extraction.

This paper proposes a multimodal pipeline leveraging audio (speech) and visual (whiteboard) to produce accurate, structured lecture notes for Banglish content.

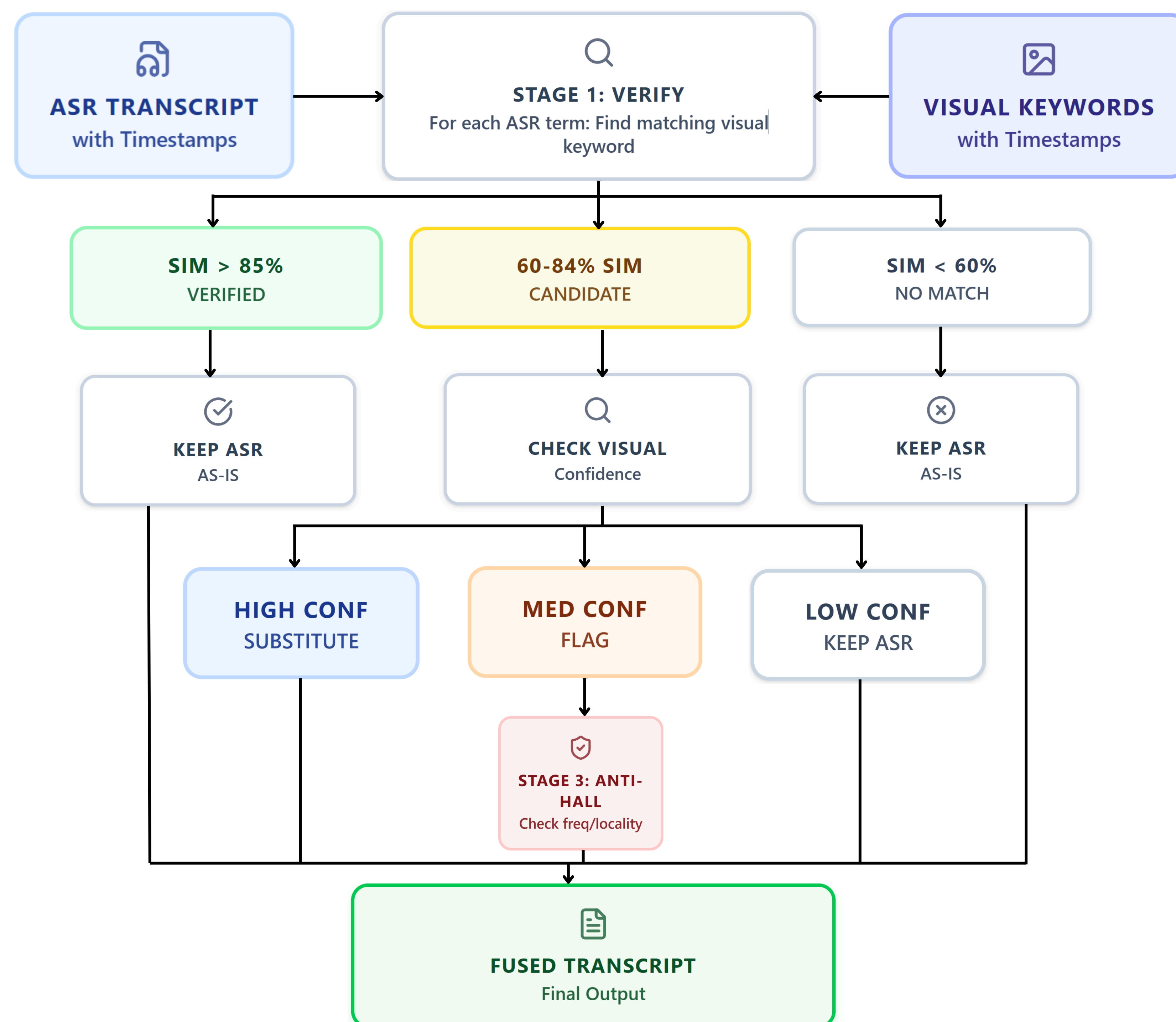
Dataset

Property	Value
Dataset type	Video(.mp4)
Total Videos	9(BanglaASR1-9)
Ground Truth Chars	~73000 total
Video type	Real Classroom Recordings
Domains	Python(5), DBMS(2), DLD(2)

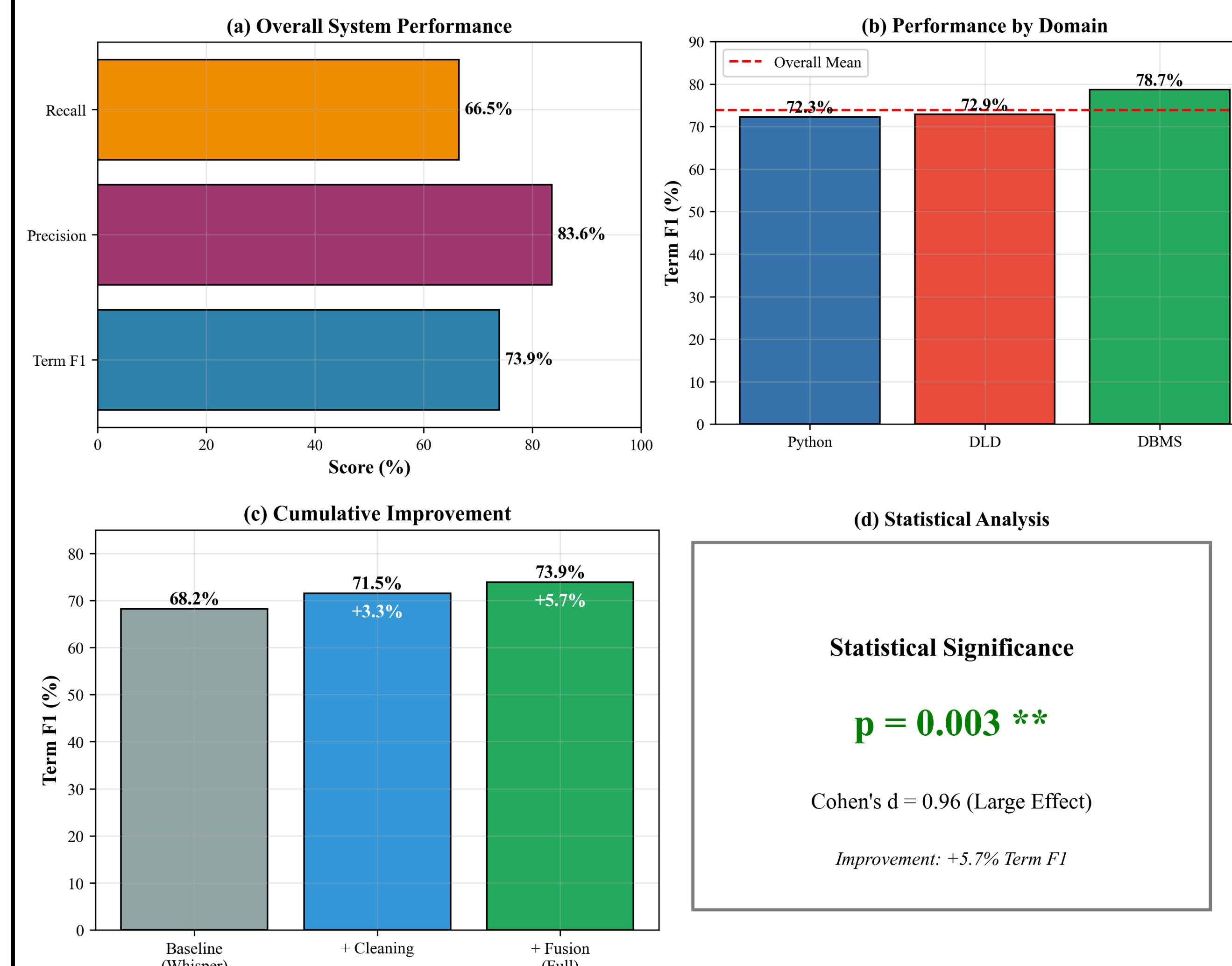
Research Pipeline and Models



Cross Modal Fusion Flowchart



Evaluation Results



A paired t-test confirmed statistically significant improvement ($t(8) = 4.23$, $p = 0.003$), with a large effect size (Cohen's $d = 0.96$). The full pipeline outperformed the baseline on all 9 videos, showing improvements in F1 scores ranging from +2.1% to +9.8%.

Contribution and Conclusion

1. **First Banglish Benchmark Dataset**: 9 manually transcribed lecture videos (~73K characters), Enables reproducible research.
2. **Term-Based Evaluation Framework**: Fuzzy matching for Romanized Banglish (F1 73.9%).
3. **Multimodal Pipeline**: Dual-ASR + VLM + LLM summarization end-to-end.
4. **Anti-Hallucination Processing**: Removes repetition loops.

Future Work

- **Frequency-Aware Cross-Modal Verification**: Detect over-representation of whiteboard terms.
- **Bengali Normalization**: Improve transliteration accuracy.
- **Temporal Alignment**: Better audio-visual synchronization.
- **Extended Dataset** specially focused on Banglish Language.

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